CONSTRUCTION ECONOMICS AND FINANCE

Semester Course Project: AI-Assisted Decision-Support Tools

Project Guidelines, Topics, Deliverables and Assessment

Item	Details
Course	CE725 - Construction Economics and Finance
Team size	5 students per team
Duration	One semester (approximately 4 weeks)
Final submission	November 01 ? (tool, reports and presentation)

1. Project Overview

Each team will build a working decision-support tool that addresses a real problem in construction economics or finance. Teams will use AI tools to research the problem, design the method, build and test the tool, and then critically reflect on what the AI did well, where it failed, and what the team learned about the subject.

1.1 Learning Outcomes

On completion of the project, students will be able to:

- Frame a construction economics or finance problem clearly and select an appropriate analytical method (NPV, IRR, life cycle costing, earned value, risk analysis and similar).
- Use AI tools productively for research, modelling and coding.
- Validate AI-generated outputs against independent manual calculations.
- Communicate results and limitations to a non-technical client or decision-maker.
- Reflect on the reliability, limits and responsible use of AI in professional practice.

1.2 Core Principle

The AI builds with the team, not for the team. Every number produced by the tool must be verified against a manual benchmark calculation. A tool that runs but has not been validated will earn limited credit, regardless of how polished it looks.

2. Team Structure and Roles

All members contribute to the tool and must be able to explain every part of it. Roles define accountability, not exclusive ownership of work. The following may be considered as a indicative list of roles which might be needed for your project.

Role	Responsibilities
Project Lead and Finance Modeller	Defines scope and timeline; owns the core economic logic, formulas and assumptions.
Data Lead	Collects, cleans and documents data such as rates, indices and project data; maintains the source list.

Role	Responsibilities
AI and Tool Developer	Builds the tool with AI-assisted coding or spreadsheets; manages versions and files.
Validation Engineer	Prepares the manual benchmark, tests edge cases and records every AI error found.
Documentation and Reflection Lead	Maintains the AI usage log, user manual, final report and presentation.

3. Suggested AI Toolkit

Teams may use any AI tools available to them. The following are **indicative.** I would be happy if you suggest any other platform. Be aware of your technology choice. Your team should reflect the implications of confidentiality, privacy and data security from the user of the tool perspective:

Purpose	Example Tools	Typical Use
Research and literature	Perplexity, NotebookLM, Consensus, Elicit	Finding methods, standards and papers; all citations must be verified
Reasoning and coding	ChatGPT, Claude, Gemini, GitHub Copilot	Formula logic, Python code, spreadsheet formulas, debugging
Build platforms	Excel with Copilot, Google Sheets with Gemini, Python (Google Colab, Streamlit)	Building the working tool
Dashboards	Power BI, Looker Studio	Visual reporting of results
No-code apps	Glide, AppSheet	Simple user-facing tools without programming
Document intelligence	Any capable LLM	Reading contracts, tenders and financial statements

4. Semester Timeline

Phase	Output
Team formation, topic selection and problem framing	One-page project charter
Literature and method study (AI-assisted); data collection plan	Method note and data source list
Manual benchmark: solve one sample case by hand or in a plain spreadsheet	Benchmark workbook
Tool development with AI assistance in iterative builds	Working prototype; mid-term demonstration in week 9
Validation, sensitivity testing and error logging	Validation report
User testing with peers or an industry practitioner; refinement	Final tool
Final report, reflection, presentation and live demonstration	Final submission

5. Project Topics

5.1 Summary of Topics

No.	Topic
1	AI-Assisted Construction Cost Estimator (BOQ Generator)
2	Real Estate Project Feasibility and Financial Appraisal Tool
3	Contractor Cash Flow and Working Capital Forecaster
4	Material Price Forecasting and Escalation Calculator
5	Monte Carlo Risk Analysis for Cost and Schedule Contingency
6	Life Cycle Cost Analysis (LCCA) Decision Tool
7	Earned Value Management (EVM) Dashboard
8	Contractor Financial Health and Prequalification Scorer
9	PPP / Infrastructure Project Financial Model
10	Bid Pricing and Markup Optimisation Tool
11	Equipment Economics: Buy vs Lease vs Rent Analyser
12	AI Contract Analyser for Financial Risk Clauses
13	Delay Claim and Variation Quantification Tool
14	Developer Funding Structure and Escrow Compliance Analyser

5.2 Topic Descriptions

Topic 1: AI-Assisted Construction Cost Estimator (BOQ Generator)

Problem: Preliminary estimates at the concept stage are slow and depend heavily on the estimator's experience.

Tool to be developed: Takes building parameters (built-up area, number of floors, structural system, specification level) and produces a preliminary bill of quantities and an element-wise cost estimate using a standard schedule of rates (e.g., CPWD DSR or the applicable state SOR).

Role of AI: Quantity logic, mapping of items to rate codes, formula and code generation, generation of the cost summary narrative.

Validation: Compare the tool's estimate for one building against a detailed estimate prepared manually or taken from a textbook or real project.

Topic 2: Real Estate Project Feasibility and Financial Appraisal Tool

Problem: Developers need to judge viability before committing to land and approvals.

Tool to be developed: Models land cost, construction cost, approval costs, sales phasing, financing and taxes; computes NPV, IRR, payback period, profit on cost and break-even selling price; includes sensitivity tables and a tornado chart.

Role of AI: Model structuring, formula checking, drafting a short investment memo from the model output.

Validation: Recompute NPV and IRR for the base case by hand or in a plain spreadsheet without AI assistance.

Topic 3: Contractor Cash Flow and Working Capital Forecaster

Problem: Profitable contractors often fail because of cash shortages rather than losses.

Tool to be developed: From a project schedule, payment terms, retention, mobilisation advance and billing cycle, generates S-curves, cash-in versus cash-out curves, the peak funding requirement and the interest cost of overdraft.

Role of AI: Converting schedules into monthly cash flows, charting code, scenario generation.

Validation: Manually compute monthly cash flows for a 6 to 12 month sample project and compare.

Topic 4: Material Price Forecasting and Escalation Calculator

Problem: Volatile prices of steel, cement, bitumen and fuel erode contractor margins on long contracts.

Tool to be developed: Uses historical price index data (e.g., Wholesale Price Index series) to build a forecasting model and includes a price variation clause calculator based on standard escalation formulas.

Role of AI: Time-series code, model selection guidance, interpretation of forecast errors.

Validation: Back-test forecasts on a held-out period and compute the price variation amount for one bill manually.

Topic 5: Monte Carlo Risk Analysis for Cost and Schedule Contingency

Problem: Contingency is commonly set as a flat percentage without evidence.

Tool to be developed: Accepts three-point estimates (optimistic, most likely, pessimistic) for cost items and activities, runs simulations, and reports P50, P80 and P90 values with histograms and cumulative curves.

Role of AI: Simulation code, choice of probability distributions, explanation of results for management.

Validation: Compare simulated means with PERT mean calculations done by hand; test with a small number of items.

Topic 6: Life Cycle Cost Analysis (LCCA) Decision Tool

Problem: Decisions based only on initial cost ignore operating, maintenance and replacement costs.

Tool to be developed: Compares alternatives such as rooftop solar versus grid power, flooring types, facade systems or HVAC options over the life of the building, including residual value, discount-rate sensitivity and optionally carbon cost.

Role of AI: Gathering typical cost and life data (to be verified), formula generation, comparison reporting.

Validation: Hand-calculate present worth for two alternatives and compare with the tool.

Topic 7: Earned Value Management (EVM) Dashboard

Problem: Project managers need early warning of cost and schedule overruns.

Tool to be developed: Takes planned versus actual progress and cost data and computes PV, EV, AC, CPI, SPI, EAC, ETC and TCPI with trend charts.

Role of AI: Dashboard building, formula checking, automatic plain-language monthly progress commentary.

Validation: Manually compute all EVM indices for two reporting periods.

Topic 8: Contractor Financial Health and Prequalification Scorer

Problem: Clients need to assess whether a bidder can financially sustain a project.

Tool to be developed: Reads balance sheets and profit and loss statements of listed construction companies, computes liquidity, leverage, profitability and working capital ratios and the Altman Z-score, and produces a prequalification score. Tested on 8 to 10 real companies.

Role of AI: Extraction of figures from annual reports, ratio formulas, scoring rationale.

Validation: Check extracted figures against the original annual reports; hand-compute ratios for two companies.

Topic 9: PPP / Infrastructure Project Financial Model

Problem: PPP projects such as toll roads, metro lines and Hybrid Annuity Model (HAM) projects depend on sound financial structuring.

Tool to be developed: Models debt-equity structure, interest during construction, revenue ramp-up, DSCR, LLCR, project IRR and equity IRR, with scenarios for traffic shortfall and cost overrun.

Role of AI: Model architecture, debt schedule logic, scenario setup, explanation of lender metrics.

Validation: Verify the debt repayment schedule and DSCR for selected years by hand.

Topic 10: Bid Pricing and Markup Optimisation Tool

Problem: Contractors must balance the probability of winning against expected profit.

Tool to be developed: Supports a bid/no-bid decision using weighted criteria and estimates the optimal markup using competitor bid history (Friedman or Gates models).

Role of AI: Analysis of past tender results, probability calculations, code generation.

Validation: Hand-compute expected profit at three markup levels for one competitor scenario.

Topic 11: Equipment Economics: Buy vs Lease vs Rent Analyser

Problem: Equipment decisions tie up significant capital and affect project cost.

Tool to be developed: Computes ownership and operating cost per hour (depreciation, interest, insurance, fuel, maintenance, operator), compares buy, lease and rent options across utilisation levels including tax effects of depreciation, and plots break-even utilisation.

Role of AI: Cost component formulas, typical data (to be verified), charting.

Validation: Hand-calculate hourly ownership and operating cost for one machine.

Topic 12: AI Contract Analyser for Financial Risk Clauses

Problem: Financial risk is buried in long contract and tender documents.

Tool to be developed: Uses an LLM to extract payment terms, retention, liquidated damages, escalation, advance and bank guarantee clauses, flags financial risks and estimates their cash flow impact.

Role of AI: Core engine of the tool; the team must document where the model misreads or invents clauses.

Validation: Manually extract clauses from two contracts and measure the tool's accuracy (correct, missed, incorrect).

Topic 13: Delay Claim and Variation Quantification Tool

Problem: Contractors often under-recover delay costs because claims are poorly quantified.

Tool to be developed: Given delay events and project data, estimates extended overheads (e.g., Hudson and Emden formulas), idle resource costs and escalation, and produces a claim summary.

Role of AI: Formula implementation and drafting of a claim narrative that students must fact-check.

Validation: Hand-compute overhead claims using both formulas for one delay scenario.

Topic 14: Developer Funding Structure and Escrow Compliance Analyser

Problem: Real estate developers must choose between funding routes while meeting regulatory requirements.

Tool to be developed: Compares equity, construction finance, customer advances and structured debt; tracks escrow requirements (e.g., the RERA requirement to keep 70% of collections in a separate account) and shows the effect on project IRR and risk.

Role of AI: Regulation summaries (to be verified against the Act and state rules), model structuring, scenario analysis.

Validation: Check escrow balances and withdrawals for a sample period by hand.

6. Final Deliverables

1. **The tool:** spreadsheet file, Streamlit application link, Colab notebook or dashboard, together with all source files.
2. **User manual (3 to 5 pages):** how to use the tool, required inputs, key assumptions and known limitations.
3. **Validation report:** the benchmark case solved manually and by the tool, a table of discrepancies, and how each was resolved.
4. **AI usage log:** record of significant prompts, what the AI produced, and whether the output was accepted, modified or rejected, with reasons. A minimum of 20 significant entries per team (format in Annexure A).
5. **Learnings and reflection report:** structured as described in Section 7.
6. **Final presentation:** 15-minute live demonstration followed by 5 minutes of questions. Any team member may be asked to explain any formula or result.

7. Structure of the Learnings and Reflection Report

The report should be 10 to 15 pages and contain the following sections:

1. **Subject learnings:** what the team now understands about construction economics and finance that it did not understand before.
2. **AI capability assessment:** where AI accelerated the work and where it was wrong, with specific examples of incorrect formulas, wrong assumptions, fabricated data or faulty code.
3. **Validation insights:** what the comparison with the manual benchmark revealed.
4. **Practical relevance:** whether a site engineer, contractor, developer or lender would use the tool, and what would need to change for real use.
5. **Individual reflections:** half a page per member on their role and personal learning.
6. **Ethics and limitations:** data reliability, over-reliance on AI, and accountability for decisions based on AI-generated numbers.

8. Assessment Scheme

Component	Weight
Problem framing and correctness of method	15%
Tool functionality and usability	20%
Validation against manual benchmark	20%
AI usage log (depth and critical judgement)	15%
Learnings and reflection report	15%
Presentation, demonstration and individual viva	15%
Total	100%

8.1 Additional Assessment Notes

- An individual viva/tool demonstration/poster presentation will be held during the final demonstration to confirm that every member understands the economics, not only the tool.
- Bonus credit may be awarded to teams that test the tool with a practising engineer, contractor or bank or industry professional and document the feedback.
- Undisclosed use of AI, or submission of unverified AI output as the team's own analysis, will be treated as academic misconduct.

Annexure A: AI Usage Log Format

Teams should maintain the log throughout the semester, not reconstruct it at the end.

No.	Date	Tool	Purpose / Prompt Summary	Output Summary	Decision (Accept / Modify / Reject)	Reason and Verification
1						
2						
3						

Annexure B: Project Charter Format

Item	Details
Team number	
Team members and roles	
Topic selected	
Problem statement (100 words)	
Intended users of the tool	
Method(s) to be applied	
Data sources	
Build platform	
Benchmark case for validation	
Key risks to completion	